5

ME1100 Thermodynamics

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What is a pure substance?

- **‘Pure substance’** is a **substance which has uniform composition and chemical aggregation** throughout the process under consideration.

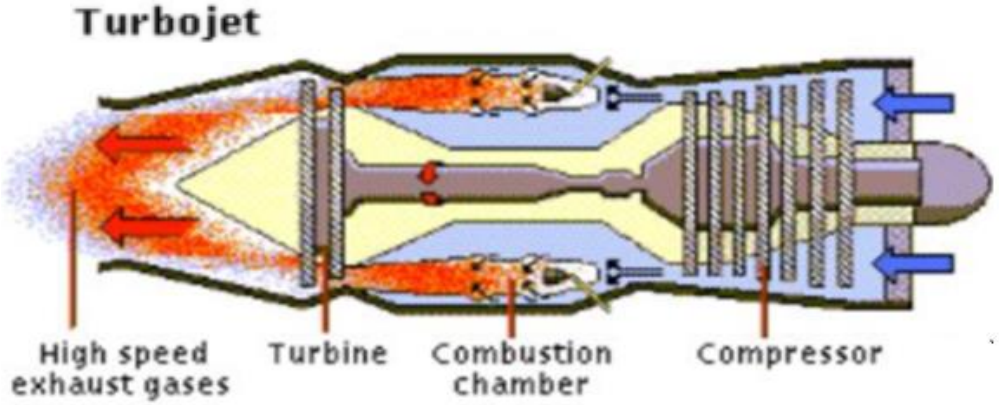
e.g. oxygen, nitrogen, water, etc.

What about the following examples:

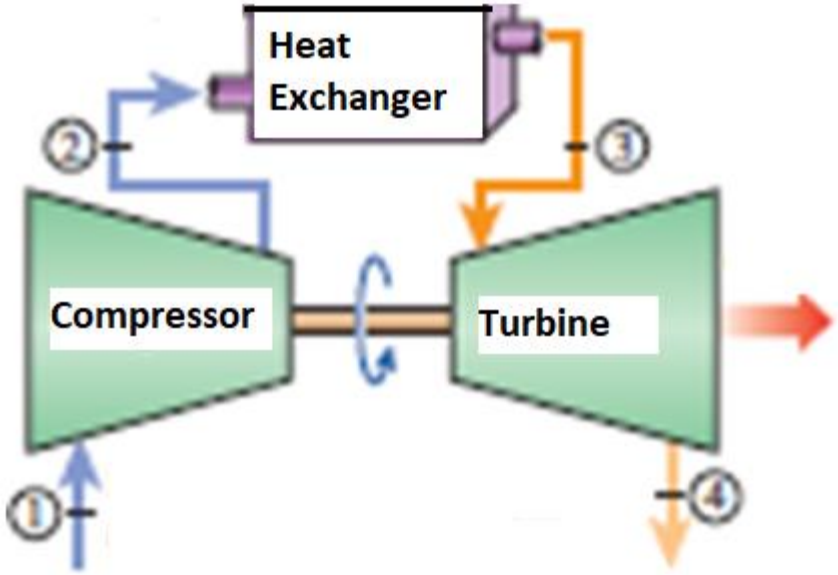
- air (which is a mixture of several gases) ? √ X
- water-steam mixture √
- combustion of oxygen-methane X

Why should we study property of pure substances?

- Gas turbine based power plant → Air (gas) in Gas Power Cycle



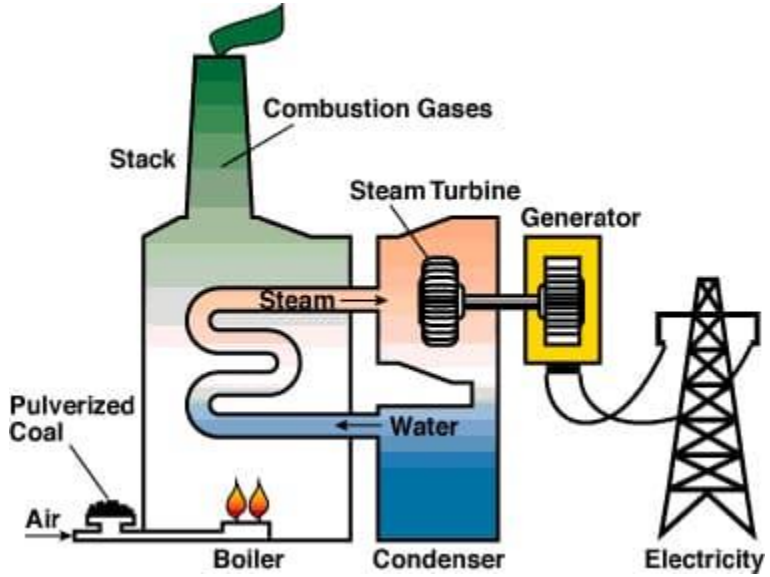
http://image.slideserve.com/242872/gas-power-cycle-jet-propulsion-technology-a-case-study-l.jpg



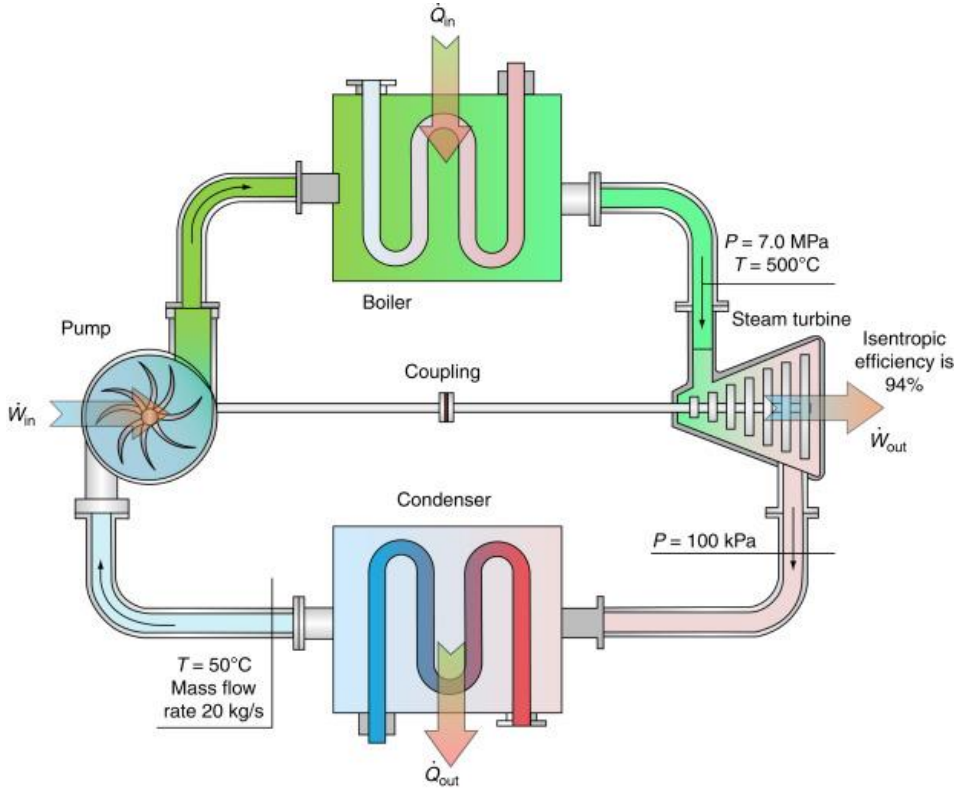
Adapted from Cengel and Boles, Thermodynamics

Why should we study property of pure substances?

- Steam is working fluid in steam power plant



https://swh-826d.kxcdn.com/wp-content/uploads/2011/06/steam-turbines.jpg

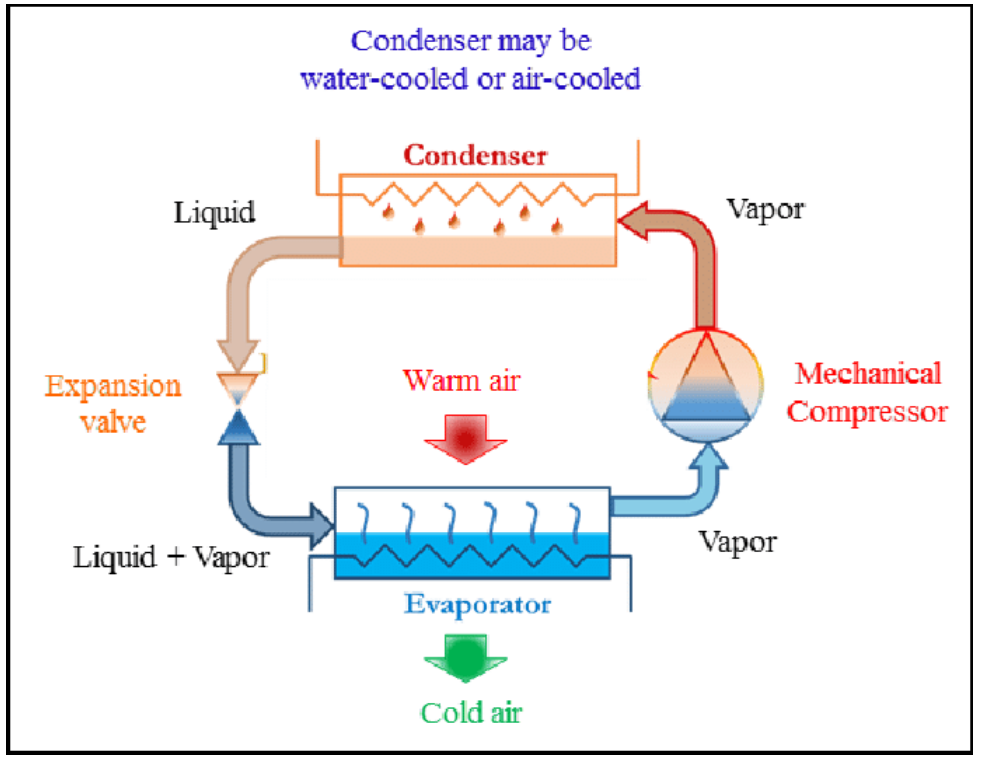


https://www.sciencedirect.com/topics/engineering/steam-rankine-cycle

Why should we study property of pure substances?

- Refrigerants like R134a are used

We find phase changes from liquid to vapour and vice versa in **steam power plant** or in **refrigeration system**



https://www.researchgate.net/figure/Schematic-diagram-of-a-typical-vapor-compression-refrigeration-cycle-17_fig1_326272160

Gibbs Phase Rule

$F=c-\varphi+2$

F: degrees of freedom
c: number of components
φ: number of phases

Gases like O_2, N_2, etc. ➡ **F=1-1+2=2**

Recall ideal gas laws: **PV=nRT**

Two-property rule :
"The state of a pure substance of given mass can be specified in terms of **two independent** properties, in the absence of effects due to gravity, motion, electricity, magnetism, elastic deformation, etc."

Liquid-Vapour System

$F=c-\varphi+2$

Water liquid and water vapour(steam) are in **thermodynamic equilibrium**

$F=1-2+2=1$

Thus, we will see that specifying temperature will fix pressure or vice versa.
e.g. we know that water boils at 100 °C under 1 atmosphere of pressure.

If, ice, water liquid and vapour are in **thermodynamic equilibrium**

$F=1-3+2=0$

Pure substances

- **Ideal gas**
- **Mixture of ideal gases**
- **Steam/ water substance**
- **R134a**

Ideal Gases & Ideal Gas Mixtures

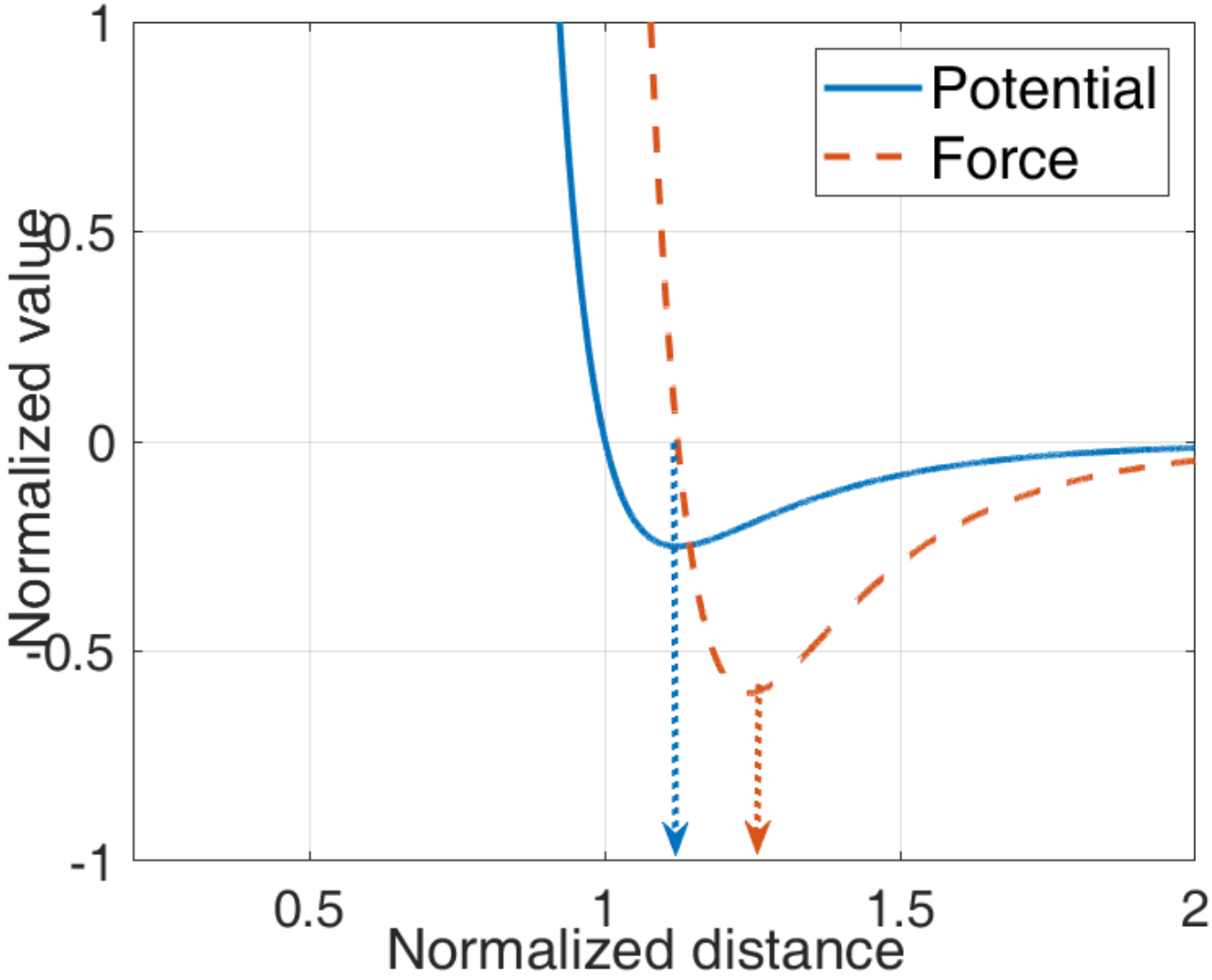
Properties of ideal gas

- Molecules are under constant random motion and collisions between all molecules are perfectly elastic

- Any gas can be treated as an ideal gas if its molecules exert negligible attractive (Van der Waals) forces on each other.

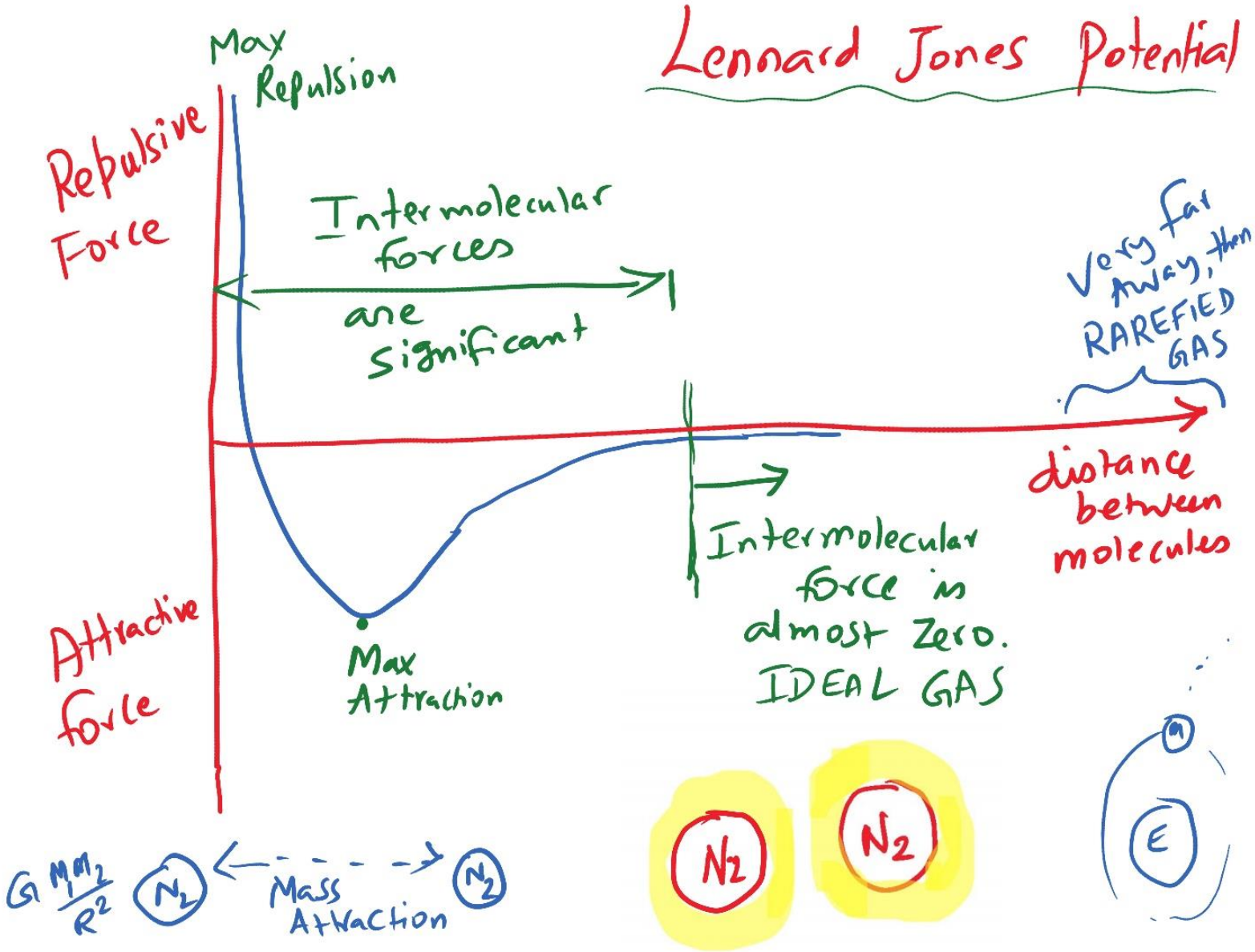
- Since this condition becomes applicable for molecules which are separated wide apart, the ideal gas assumption becomes valid as $V \to \infty$ and $p \to 0$.

Intermolecular Potential



$$\vec{F} = -\frac{d\Phi}{dr}$$

Lennard Jones Potential



Boyle's and Charles' laws

An ideal gas obeys the Boyle's and Charles' laws

Boyle's Law

At constant temperature, the volume of a given mass of gas is inversely proportional to its pressure

$$V \propto {}^{1}/_{p}$$

Charles' law

At constant pressure, the volume of a given mass of gas is directly proportional to its temperature

$$V \propto T$$

Boyle's and Charles' laws

Combining $V \propto {}^{1}/_{p}$ **and** $V \propto T$

$$\frac{p\,V}{T}=\text{constant}$$

$$pV=n\bar{R}T$$

in K

$$pV=mRT$$

m mass of the gas

Universal gas constant

$$\bar{R} = 8314\,\frac{J}{kmol}\,K$$

$$R = \frac{\bar{R}}{M}$$

Specific gas constant

M molecular weight of the gas (kg/kmol)

Ideal Gas

In general for any pure substance internal energy (U) depends on two properties specifying the state.

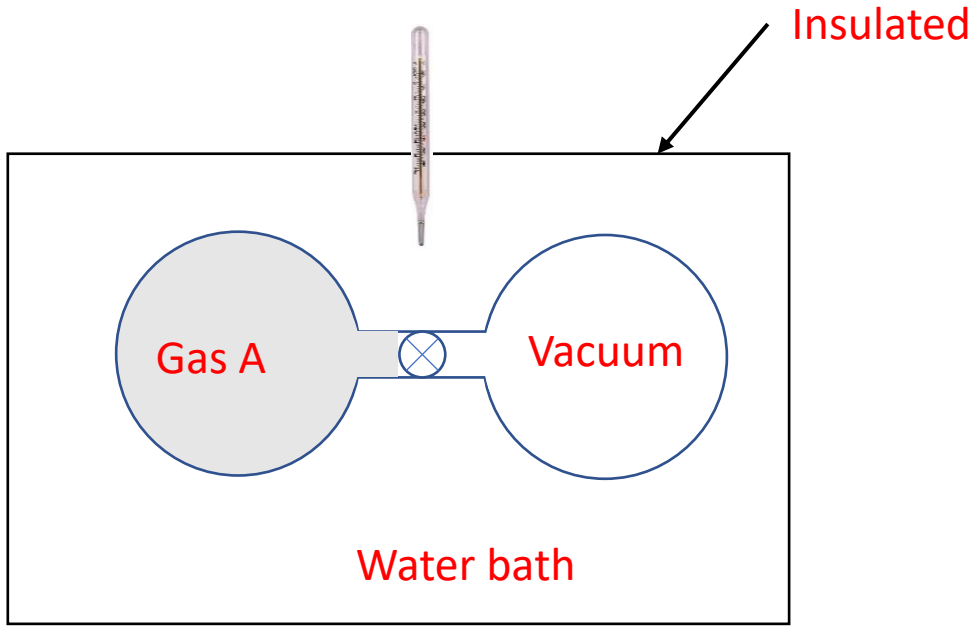
For low density gases it primarily depends on depends on temperature and much less on pressure or volume

$$U = U(T)$$ **or u=u(T)**

$$h = u + pv$$
$$h = u + RT$$

$$h = h(T)$$

Joule's Experiment



The temperature was constant

First Law of Thermodynamics

$$Q - W = \Delta E = \Delta U$$

$$Q_{water} + Q_{Gas} - W_{water} - W_{Gas} = \Delta U_{water} + \Delta U_{Gas}$$

(Q_{water}, Q_{Gas}, W_{water}, W_{Gas}, ΔU_{water} = 0)

Thus, Joule's experiments proved that internal energy of an ideal gas depends on temperature alone

$$\Delta U_{Gas} = 0$$

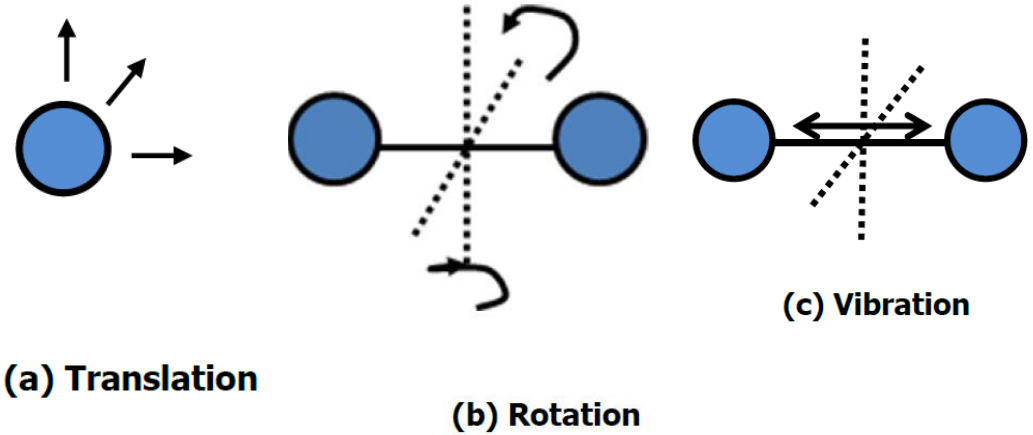
$$U_1 = U_2$$

$$U = U(T)$$

Ideal Gas

Internal energy of a gas depends on

Translational KE
Rotational KE
Vibrational KE
} of molecules



$$U = \frac{1}{2} k_B T$$

k_B – Boltzmann Constant

$1.38 \text{ X } 10^{-26}$ kJ/K

$$k_B = \frac{\bar{R}}{N_A}$$

N_A – Avogadro Number

$6.023 \text{ X } 10^{26}$ molecules/kmol

Ideal Gas

Total degrees of freedom - D

Internal energy per molecule - $\frac{1}{2}k_B T D$

Molar internal energy (internal energy per kmol of gas)

$$\bar{U} = \frac{1}{2}k_B T D N_A$$

$$= \frac{D}{2}\bar{R}T$$

Molar enthalpy $= \bar{H} = \frac{D+2}{2}\bar{R}T$

Ideal Gas

$\overline{U}$ and $\overline{H}$ can be expressed in terms of molar specific heats

$$d\overline{U} = \frac{D}{2}\overline{R}dT$$

$$\overline{C_v}dT = \frac{D}{2}\overline{R}dT \text{ or } \overline{C_v} = \frac{D}{2}\overline{R}$$

$$d\overline{H} = \frac{D+2}{2}\overline{R}dT$$

$$\overline{Cp}dT = \frac{D+2}{2}\overline{R}dT \text{ or } \overline{C_p} = \frac{D+2}{2}\overline{R}$$

$\overline{U}$, $\overline{H}$ - *J/kmol*

Ideal Gas

In terms of specific heats

$$\mathrm{u} = \frac{\bar{U}}{M} = \frac{D}{2}\frac{\bar{R}}{M}T$$

$$C_v dT = \frac{D}{2}RdT \text{ or } C_v = \frac{D}{2}R$$

$$h = \frac{\bar{H}}{M} = \frac{D+2}{2}\frac{\bar{R}}{M}T$$

$$C_p dT = \frac{D+2}{2}RdT \text{ or } C_p = \frac{D+2}{2}R$$

Ideal Gas

Ratio of specific heats

$$\gamma = \frac{C_p}{C_v} = \frac{D+2}{D}$$

For monoatomic gases D=3 γ =1.67

For diatomic gases D=5 γ =1.4

$$dh = du + RdT$$
$$C_p dT = CvdT + RdT$$
$$C_p = Cv + R$$
$$\gamma = \frac{C_p}{C_v} \quad C_p = \frac{\gamma R}{\gamma - 1}$$

Let us apply the above relations to obtaining the specific heat values for nitrogen at low temperatures.

The molecular weight of nitrogen $M = 28$ kg/kmol.

Then the specific gas constant $R = 8314$ (J/kmol K) /28 (kg/kmol)
$= 296.93$ J/kg K.

Since the specific heat ratio $\gamma = 1.4$ (for degrees of freedom $= 5$),

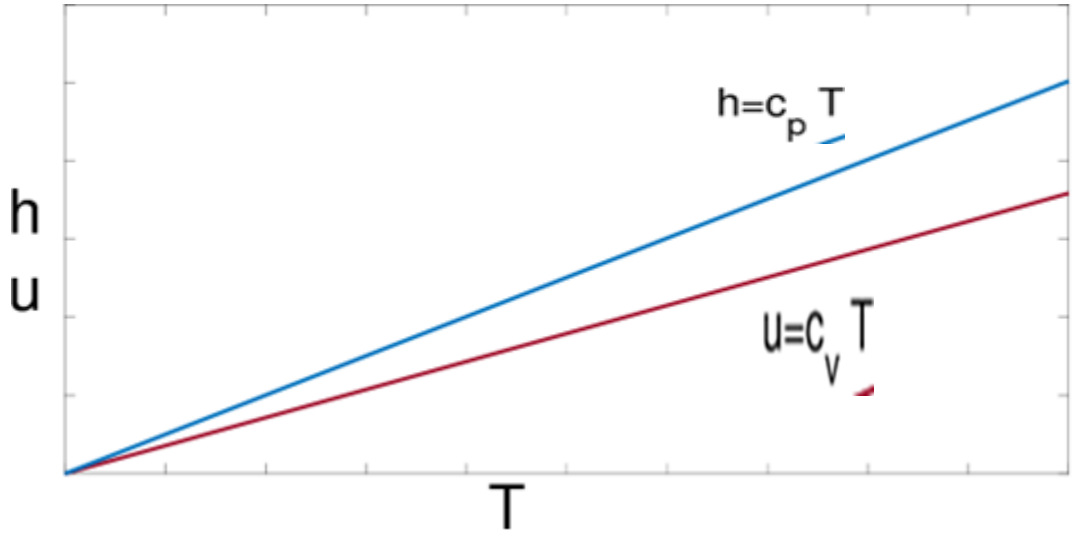
$C_v = 296.93/(1.4 - 1.0) = 742.32$ J/kg K

$C_p = \gamma C_v = 1.4 \times 742.32 = 1039.25$ J/kg K

Ideal Gas

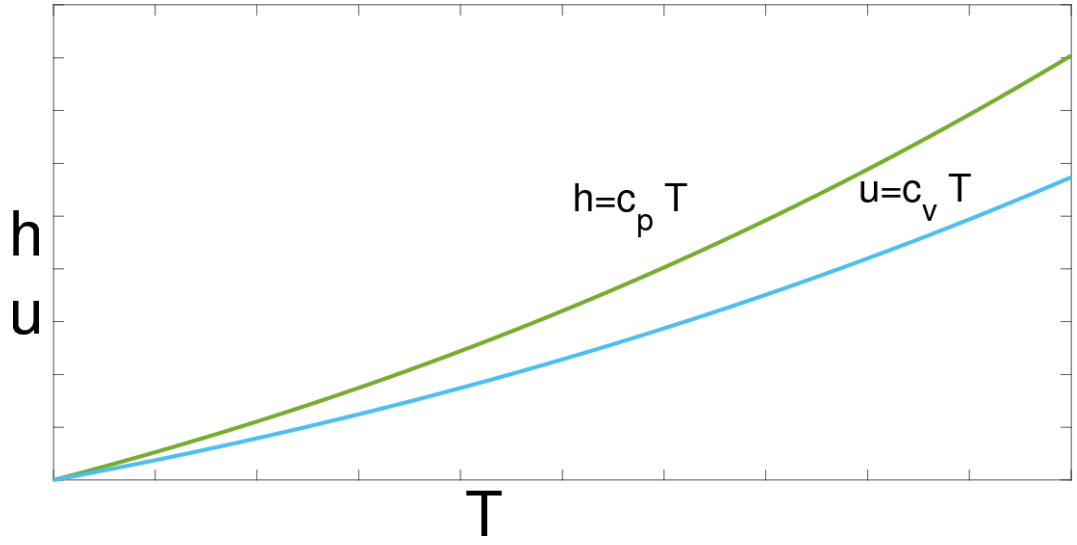
- Number of degrees of freedom may change with temperature as vibrational mode gets excited at higher temperature.

- With increase in temperature fraction of molecules with vibrational mode increases, thus the value of D increases in expressions of $\gamma,\ C_v\ and\ C_p$



- For perfect gases, specific heats are constant, so the enthalpy (H) and internal energy (U) are linear functions of temperature

- For semi-perfect gases H and U are nonlinear functions of temperature



Ideal Gas Mixture

Mixture consists of different species

Composition of mixture can be expressed in terms of mass fraction or mole fraction

Similarly properties of mixture can be expressed in terms of mass based or mole based properties of individual components

Consider a mixture of N constituents

Number of moles of each species in mixture $n_1, n_2, \ldots\ldots n_N$

Mass of each species $m_1, m_2, \ldots\ldots m_N$

Total number of moles $n=n_1+ n_2+\ldots\ldots +n_N == \sum_{i=1}^{i=N} n_i$

Total mass $m=m_1+ m_2+\ldots\ldots +m_N == \sum_{i=1}^{i=N} m_i$

Ideal Gas Mixture

Mole fraction = $x_i = n_i / \sum_{i=1}^{i=N} n_i = n_i/n$

Mass fraction = $y_i = m_i / \sum_{i=1}^{i=N} m_i = m_i/m$

$$\sum_{i=1}^{i=N} x_i = 1 \qquad \sum_{i=1}^{i=N} y_i = 1$$

Conversion of Mole fraction to Mass fraction

Consider a mixture of 1 kmol

Mole fraction of each component: $x_1, x_2, \ldots\ldots x_n$

Mass of each component

$m_1 = x_1M_1,\ m_2 = x_2M_2, \ldots\ldots\ldots m_N = x_NM_N \qquad m = \sum_{i=1}^{i=N} m_i = \sum_{i=1}^{i=N} x_iM_i$

$$y_i = \frac{m_i}{\sum_{j=1}^{N} x_jM_j} = \frac{x_iM_i}{\sum_{j=1}^{N} x_jM_j} = \frac{x_iM_i}{M}$$

Ideal Gas Mixture

Conversion of Mass fraction to Mole fraction

Consider 1 kg of mixture

Mass fraction of each component $y_1, y_2, \ldots\ldots y_n$

Number of moles of each component

$n_1 = y_1 / M_1, n_2 = y_2 / M_2, \ldots\ldots\ldots n_N = y_N / M_N,$

$$n = \sum_{i=1}^{i=N} n_i = \sum_{i=1}^{i=N} y_i / M_i$$

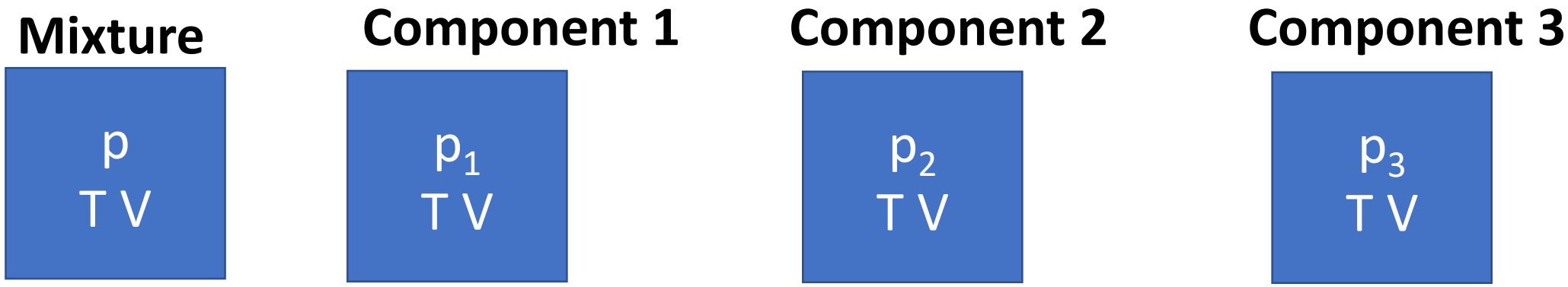
$$x_i = \frac{n_i}{n} = \frac{y_i / M_i}{\sum_{j=1}^{N} y_j / M_j}$$

Ideal Gas Mixture

Dalton's law of partial pressure

Each component in a mixture can be assumed to exert partial pressure at the mixture temperature and volume such that sum of the all partial pressures is equal to the total pressure of the mixture

$$\sum_{i=1}^{i=N} p_i = p$$

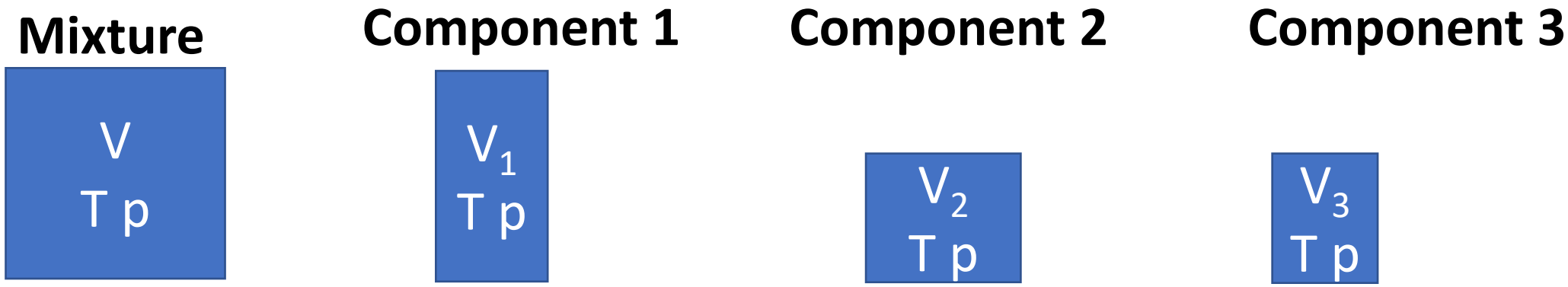


Ideal Gas Mixture

Amagat's law of partial volume

Each component in a mixture can be assumed to occupy partial volume at the mixture pressure and temperature such that sum of the all partial volumes is equal to the total volume of the mixture

$$\sum_{i=1}^{i=N} V_i = V$$



Ideal Gas Mixture

From Dalton's law

$$p_1 V = n_1 \bar{R} T$$
$$p_2 V = n_2 \bar{R} T$$
$$\text{----------}$$
$$\text{----------}$$
$$p_i V = n_i \bar{R} T$$

For mixture

$$pV = n\bar{R}T, \quad \frac{p_i}{p} = \frac{n_i}{n} = x_i$$

From Amagat's law

$$pV_1 = n_1 \bar{R} T$$
$$pV_2 = n_2 \bar{R} T$$
$$\text{----------}$$
$$\text{----------}$$
$$pV_i = n_i \bar{R} T$$

For mixture

$$pV = n\bar{R}T$$
$$\frac{V_i}{V} = \frac{n_i}{n} = x_i$$

Ideal Gas Mixture

Thus values of partial pressure or volume needed to describe the state of a component in an ideal gas mixture can be determined in terms of mole fractions.

Properties of the mixture can be evaluated from the component level by mass basis or mole basis approach

Mass basis

Internal energy

$U = mu = m_1u_1 + m_2u_2 + \cdots + m_Nu_N$

$= \sum_{i=1}^{i=N} m_i u_i$

$u = U/m = \sum_{i=1}^{i=N} \frac{m_i}{m} u_i = \sum_{i=1}^{i=N} y_i u_i$

Enthalpy

$H = mh = m_1h_1 + m_2h_2 + \cdots + m_Nh_N$

$= \sum_{i=1}^{i=N} m_i h_i$

$h = H/m = \sum_{i=1}^{i=N} \frac{m_i}{m} h_i = \sum_{i=1}^{i=N} y_i h_i$

$v = V/m = \sum_{i=1}^{i=N} \frac{m_i}{m} v_i = \sum_{i=1}^{i=N} y_i v_i$

Ideal Gas Mixture

Molar basis

Internal energy

$$U = n\bar{U} = n_1\bar{U}_1 + n_2\bar{U}_2 + \cdots + n_N\bar{U}_N$$

$$\bar{U} = U/n = \sum_{i=1}^{i=N} \frac{n_i}{n}\bar{U}_i = \sum_{i=1}^{i=N} x_i\bar{U}_i$$

Enthalpy

$$H = n\bar{H} = n_1\bar{H}_1 + n_2\bar{H}_2 + \cdots + n_N\bar{H}_N$$

$$\bar{H} = H/n = \sum_{i=1}^{i=N} \frac{n_i}{n}\bar{H}_i = \sum_{i=1}^{i=N} x_i\bar{H}_i$$

$$\bar{V} = V/n = \sum_{i=1}^{i=N} \frac{n_i}{n}\bar{V}_i = \sum_{i=1}^{i=N} x_i\bar{V}_i$$

Thus specific properties of an ideal gas mixture can be expressed as product of mass fraction and specific properties of each species summed over all species.

Similarly molar properties of a mixture can be expressed as product of mole fraction and the molar property of the component species summed over all species of the mixture.

Gas constant of a mixture

$$pV = mRT$$

$$pV = mR_{mix}T$$

$$R_{mix} = \frac{pV}{T}\frac{1}{m} = \frac{n\bar{R}}{m} = \frac{n}{m}\bar{R} = \bar{R}/M_{mix}$$

$$R_{mix} = \frac{n}{m}\bar{R} = (n_A + n_B)\frac{\bar{R}}{m}$$

$$R_{mix} = (m_A R_A + m_B R_B)\frac{1}{m} = y_A R_A + y_B R_B$$

Example

Consider an ideal gas mixture of 1 kmol of hydrogen and 1 kmol of nitrogen. Find the mass fractions and mole fractions of hydrogen and nitrogen. Also find the molecular weight, specific gas constant R and the specific heats C_p and C_v of the mixture. If this mixture is heated from an initial temperature of 300 K to the final temperature of 400K at constant pressure, find: Δu, Δh, Q and W for the process.

Molecular weight of hydrogen = 2 kg/ kmol

Molecular weight of nitrogen = 28 kg/ kmol

Total mass (m) of the mixture = 1 kmol x 2 kg/kmol + 1 kmol x 28 kg/kmol = 30 kg

Molecular weight of the mixture, M = 30 kg/ 2 kmols = 15 kg/kmol

Mass fraction of hydrogen, y_{H2} = 2 kg/ 30 kg = 1/15

Mass fraction of nitrogen, y_{N2} = 28 kg/ 30 kg = 14/15

Mole fraction of hydrogen, x_{H2} = 1 kmol/ 2 kmols = 0.5

Mole fraction of nitrogen, x_{N2} = 1 kmol/ 2 kmols = 0.5

Specific gas constant R = 8314/15 = 554.27 J/kg K

Since both hydrogen and nitrogen are diatomic gases, their specific heat ratio = 1.4 and so the specific heat ratio for the mixture γ = 1.4

Specific heat at constant volume for the mixture $C_v = R/(\gamma-1)$ = 554.27/0.4 = 1385.67 J/kg K. The value of C_p = 1.4 x 1385.67 = 1939.93 J/kg K

The change in specific internal energy $\Delta u = C_v \times \Delta T$ = 1385.67 x (400 – 300) = 138567 J/kg = 138.567 kJ/kg

The change in specific enthalpy $\Delta h = C_p \times \Delta T$ = 1939.93 x (400 – 300) = 193.993 kJ/kg

For the constant pressure process, Q = ΔH = 30 kg x 193.933 = 5818 kJ.

Change in internal energy ΔU = 30 kg x 138.567 = 4157 kJ.

Work, W = Q - ΔU = 5818 – 4157 = 1661 kJ.

Thank You

Tutorial8– problem no. 8

One kilogram of Argon (M = 40, C_v = 392 J/kg K) at 300 K, 100 kPa is contained in a rigid tank connected by a valve to another rigid tank containing 0.8 kg of O_2 at 400 K, 500 kPa. The valve is opened and the gases are allowed to mix until they reach an equilibrium temperature of 360 K. Determine (a) the volume of each tank (b) final pressure and (c) the heat interaction.

Given:

m_{Ar} =1 kg, MW_{Ar} =40 kg/kmol, $C_{v,Ar}$ =392 J/kg.K, m_{O2} =0.8 kg, MW_{O2} =32 kg/kmol, $C_{v,O2}$ =696 J/kg.K,

$T_{Ar,1}$ =300 K, $p_{Ar,1}$ =100 kPa, $T_{O2,1}$ =400 K, $p_{O2,1}$ =500 kPa, T_f = 360 K.

$\mathbf{V_{Ar,1}}$ =1×(8314/40)×300/100000=**0.62355 m^3**.

$\mathbf{V_{O2,1}}$ =0.8×(8314/32)×400/500000=**0.16628 m^3**.

Final volume, V_f = 0.62355 + 0.16628 = 0.78983 m^3.

MW_{mix} =1.0/[{1/(1.8×40)} +{0.8/(1.8×32)}]=36 kg/kmol.

R_{mix} = 8314/36=230.94 J/kg.K.

$C_{v,mix}$ =(1/1.8)×392+(0.8/1.8)×696=572.11 J/kg.K.

Using equation of state for mixture, p_f ×0.78983=(1+0.8)×230.94×360==>**p_f =189470 Pa**.

Taking both tanks as system, Q-W=ΔU ==> Q=ΔU, since W=0.

Q = 1.8×572.11×360 - [(1×392×300)+(0.8×696×400)] = **30407.28 J**.

5. 1 kg of air undergoes a cyclic process 1-2-3-4. Process 1-2 and 3-4 are adiabatic processes governed by pv^{γ} = constant, where p is the pressure and v is the specific volume and processes 2-3 and 4-1 are constant pressure processes. The temperatures T_1 and T_3 are 300 K and 1200 K respectively and the pressure ratio p_2/p_1= 10. Determine (i) T_2 and T_4 and (ii) the heat and the work interaction for each process.

Solution:

Given: m = 1 kg, T_1 =300 K, T_3 =1200 K, p_2/p_1 = 10, $Q_{1-2} = Q_{3-4} = 0$, $p_2 = p_3$ and $p_4 = p_1$

For process 1-2: $p_1v_1^{\gamma} = p_2v_2^{\gamma}$. Using pv = RT, v = RT/p,

Thus, $p_1^{1-\gamma}(RT_1)^{\gamma} = p_2^{1-\gamma}(RT_2)^{\gamma} \Rightarrow T_2/T_1 = (p_2/p_1)^{[\gamma-1]/\gamma}$

Similarly, for process 3-4, $T_4/T_3 = (p_4/p_3)^{[\gamma-1]/\gamma}$

For air, taking $\gamma = 1.4$, $T_2 = 300\times(10)^{[0.4/1.4]} = 579.2$ K.

We have, $p_4/p_3 = p_1/p_2 = 0.1$. Thus, $T_4 = 1200\times(0.1)^{[0.4/1.4]} = 621.54$ K.

For 1-2: $Q=0$, $W_{1-2} = -mC_v(T_2-T_1)$.

Molecular weight of air is 28.8 kg/kmol,

Thus, $R=8314/28.8= 288.68$ J/kg.K and $C_v = R/(\gamma-1)=721.7$ J/kg.K.

$W_{1-2} = -1\times721.7\times(579.2-300)= -201499$ J.

For 2-3: $W_{2-3} = p_2(V_3-V_2)= p_3V_3 - p_2V_2$ ($p_3= p_2$) $= mR(T_3-T_2)$

$= 1\times288.68\times(1200-579.2)=179212.5$ J.

$Q_{2-3} = mC_p(T_3-T_2)$,

$C_p = \gamma R/(\gamma-1)=1010.38$ J/kg.K. **$Q_{2-3} = 627243.9$ J.**

For 3-4:

$Q=0$, $W_{3-4} = -mC_v(T_4-T_3)$.

$= -1\times721.7\times(621.54-1200) = 417474.5$ J.

For 4-1:

$W_{4-1} = p_1(V_1-V_4)= p_1V_1- p_4V_4$ ($p_4= p_1$) $= mR(T_1-T_4)$

$= 1\times288.68\times(300-621.54)= -92822.2$ J.

$Q_{4-1} =mC_p(T_1-T_4)= -324877.6$ J.